Jeffrey Morais

Scientific Software Engineer | High-Performance Computing • Distributed Systems

Website | Email | LinkedIn | GitHub

Skills

Languages: Python, Julia, Bash, C/C++ (working knowledge)

Scientific Computing: NumPy, SciPy, PyTorch, Tensor Networks, Qiskit, PennyLane

Infrastructure: Linux, Git, CI/CD, HPC Clusters, pytest

Education

2024-2026 **M.Sc. Physics**, *University of Victoria*, Victoria, Canada

- Thesis: High-performance simulation of thermalization dynamics in quantum many-body systems, analyzing large-scale computational models of complex physical systems.
- NSERC CREATE Quantum Computing Program Fellow

2019-2023 **B.Sc. Honours Physics**, *McGill University*, Montréal, Canada

- Published honours thesis on computational topology, implementing numerical analyses of compact geometric structures and **high-dimensional state spaces** (MSURJ).

Experience

2024-Present **Quantum Algorithms Researcher**, *University of Victoria*, Victoria, Canada

- Architected Python simulation pipelines spanning 15 quantum many-body models, **scaling computational workloads across distributed Linux HPC clusters through optimized numerical workflows**.
- Engineered a high-throughput benchmarking pipeline for large-scale time-evolution analysis across broad parameter regimes.
- Analyzed large-scale simulation data to quantify thermalization-time scaling, identifying operational bounds relevant to **coherence preservation in quantum hardware**.

2023-Present **Quantum Software Lead**, *BTQ*, Vancouver, Canada

- Architected Léonne in Python, building **modular, low-latency consensus networks** leveraging topological data analysis for distributed verification.
- Engineered QLDPC, an interactive Python toolkit integrating 3D circuit construction, **efficient belief propagation decoding, and Qiskit integration**.
- Developed QRiNG, a deployable protocol for verifiable random number generation, integrating quantum key distribution into distributed consensus infrastructure.

2023-2024 **Quantum Machine Learning Researcher**, *Fudan University*, Shanghai, China

- Built TQNN, a **pip-installable Python toolkit** for noise-resilient classification architectures, establishing **rigorous CI/CD pipelines and automated testing suites**.
- Implemented gradient-free classification algorithms, improving robustness against optimization failures in variational machine learning models.
- Developed three interactive, modular GUIs for **real-time tensor network simulation**, streamlining data visualization and analysis workflows.

2022 **Computational Physics Researcher**, *McGill University*, Montréal, Canada

- **Increased weak-signal extraction efficiency by 1.7x** in noisy environments by optimizing wavelet and matched-filter algorithms in Python.
- Developed automated classification pipelines that identified target waveforms with **45% greater accuracy than baseline computational methods**.

2019-2020 **Computational Physics Researcher**, *Vanier College*, Montréal, Canada

- Developed numerical solvers for non-linear PDEs using **Crank-Nicolson methods**, generating predictive models within dynamic computational frameworks.
- Implemented RNN-driven simulations to model multi-dimensional trajectories in real-time, **adapting efficiently to arbitrary mathematical landscapes** in Python.